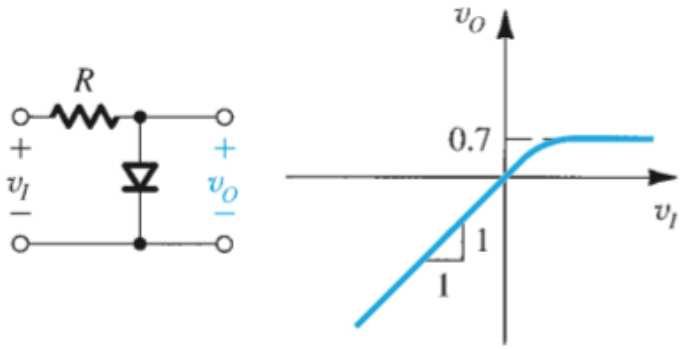
Diode Applications

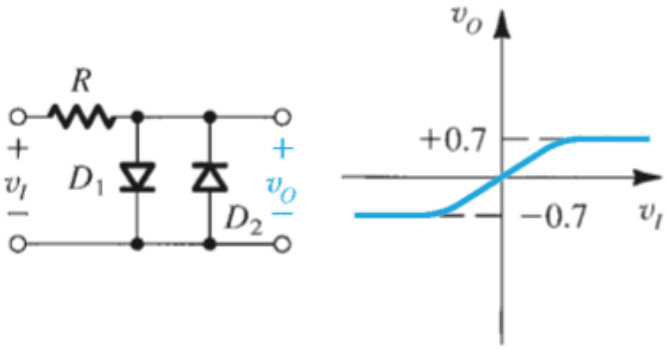
- Signal Shaping.
- Diode Types.

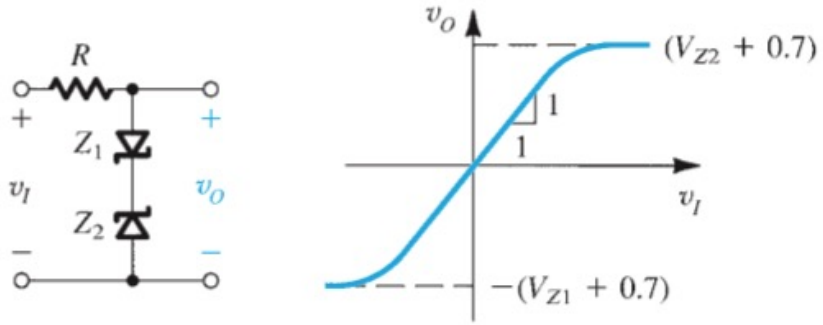
Other Diode Applications

Diode circuits work very well as waveform shaping circuits.

Clamping/clipping/limiting circuits



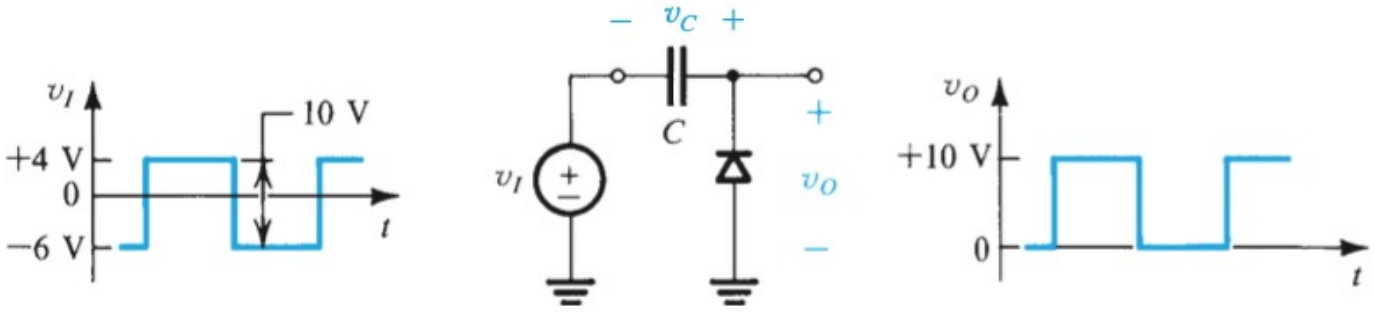


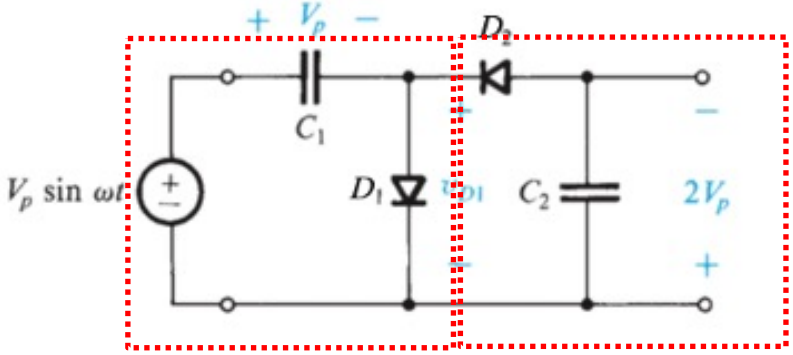


Other Diode Applications

Diode circuits work very well as waveform shaping circuits.

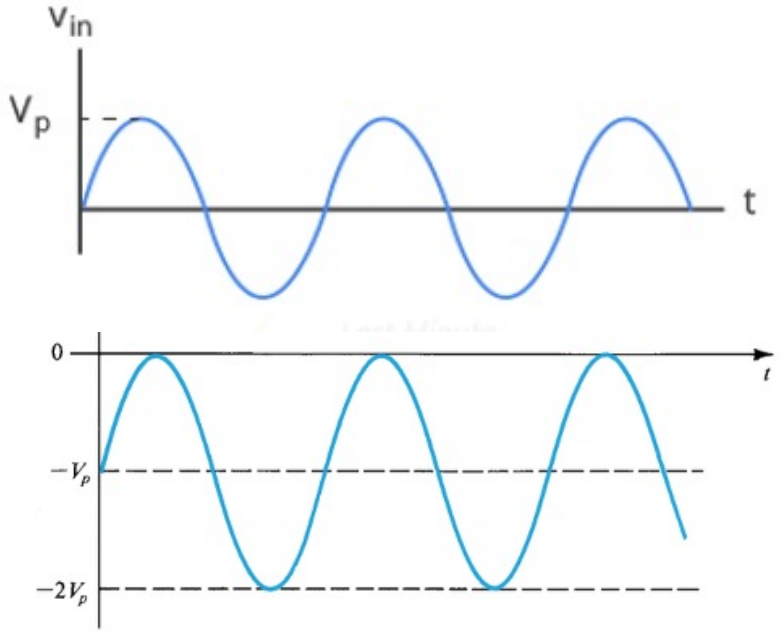
Shape forming circuits





Shift down the input signal

Peak detector



Diode Types

Regular (switching)

Zener (regulator)

Schottky diodes

Gunn/IMPATT/BARRITT

Tunneling diodes

Photodiodes/PIN/APD

Laser diodes/LED

Thyristors

Field Effect Transistors

- Introduction.
- The MOSFET.
- Current in MOSFETs and Operation Modes.
- DC Analysis (Biasing).

Introduction

From diodes to transistors

Transistors have an extra connection (3-terminal devices)

Basic applications of transistors: 1) switching 2) signal manipulation

Transistor Basic Types: FET and BJT

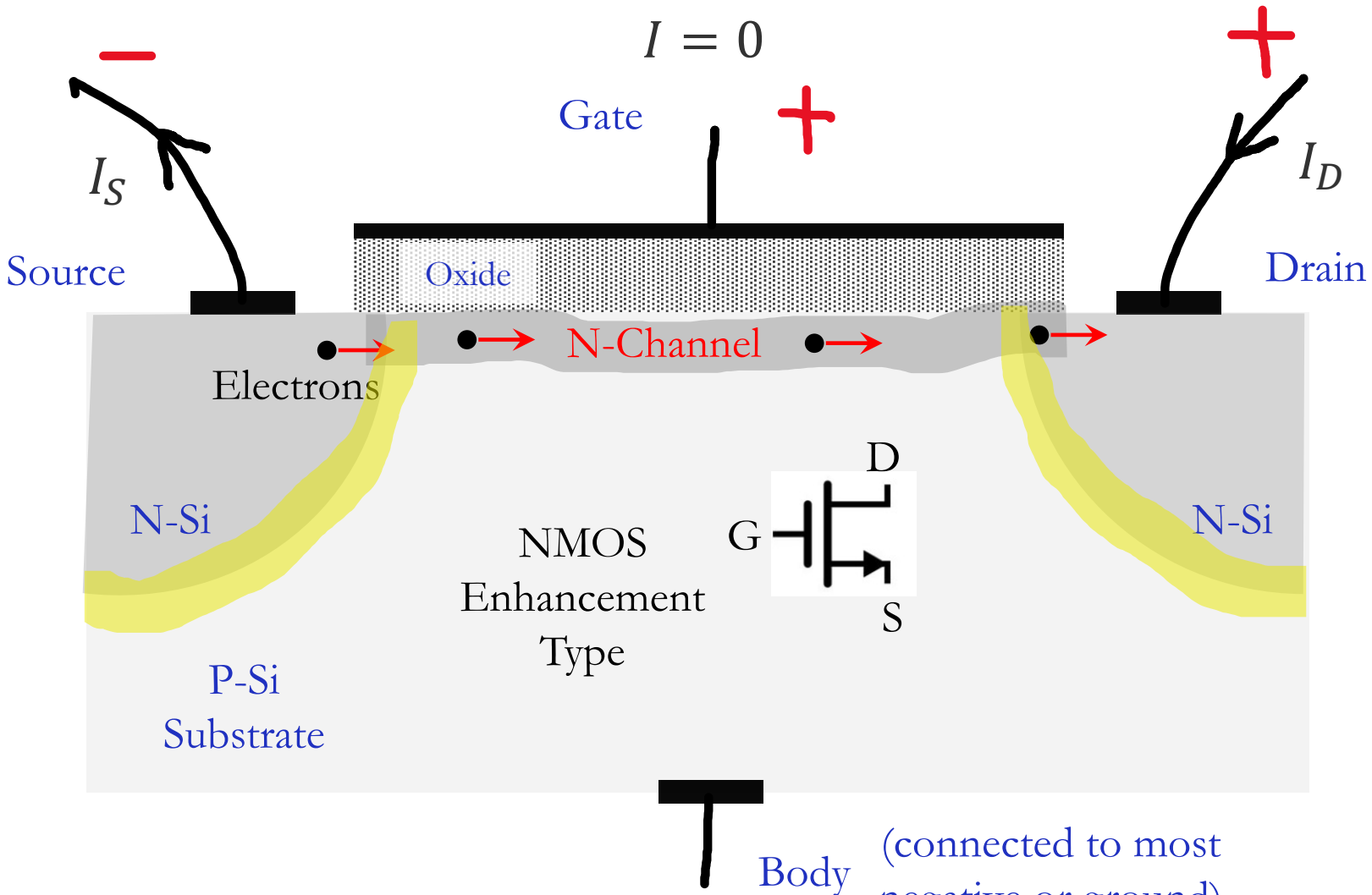
The FET family: JFET, MESFET, MOSFET, CMOS

MOSFET come with two flavors: NMOS and PMOS

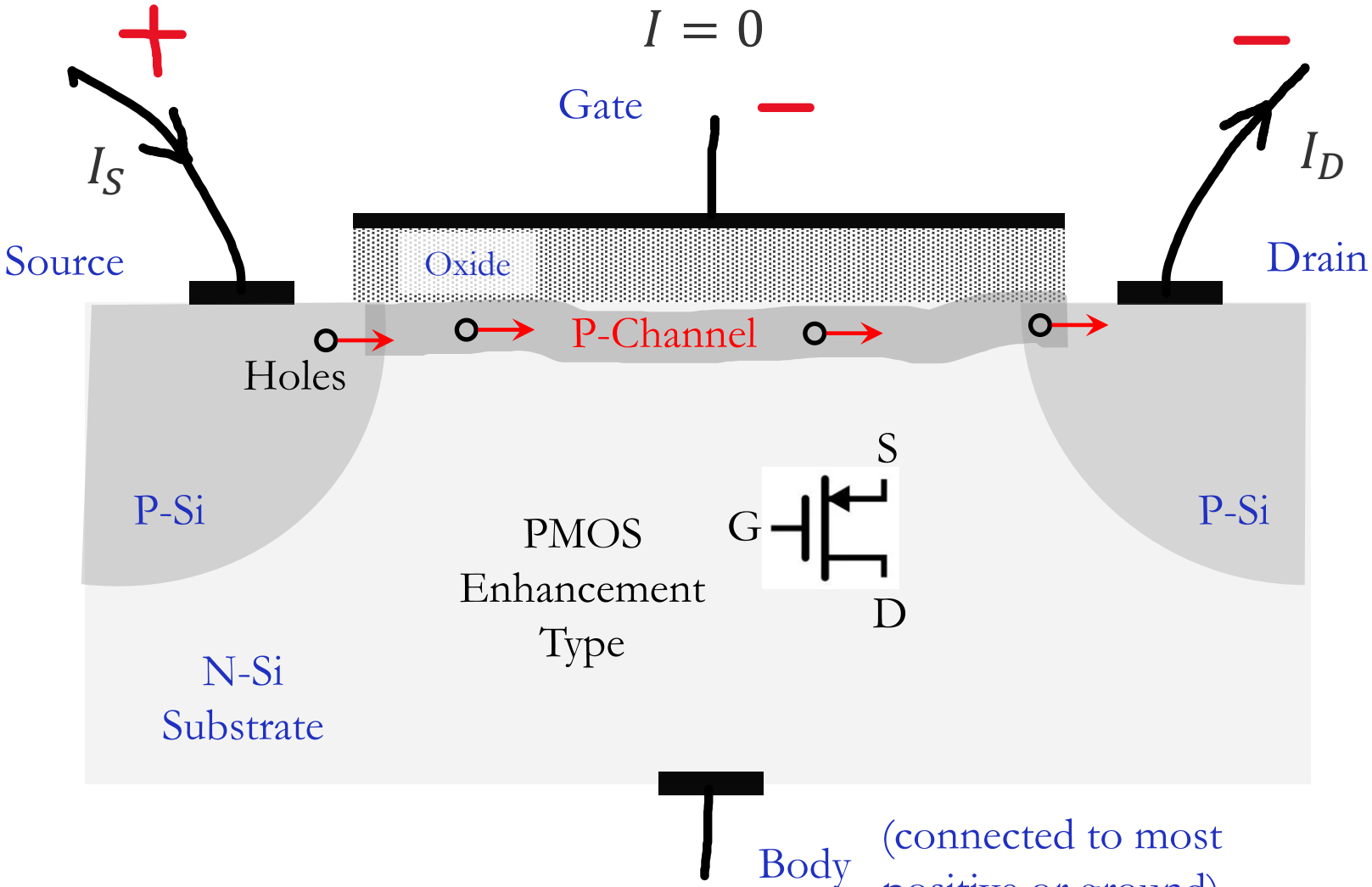
CMOS is the combination of NMOS and PMOS

MOSFET= Metal-Oxide-Semiconductor-Field-Effect-Transistor

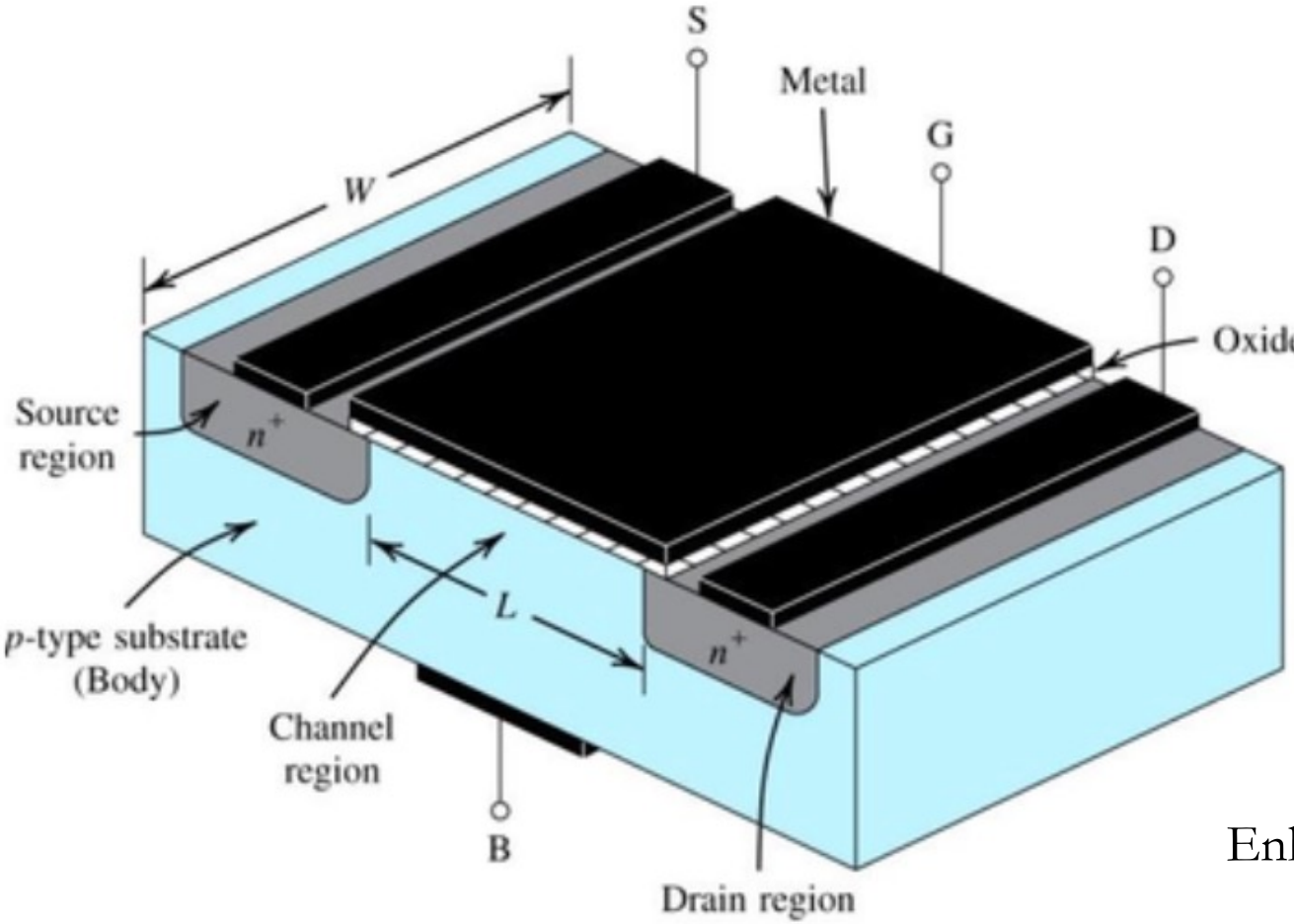
The MOSFET Structure



The MOSFET Structure



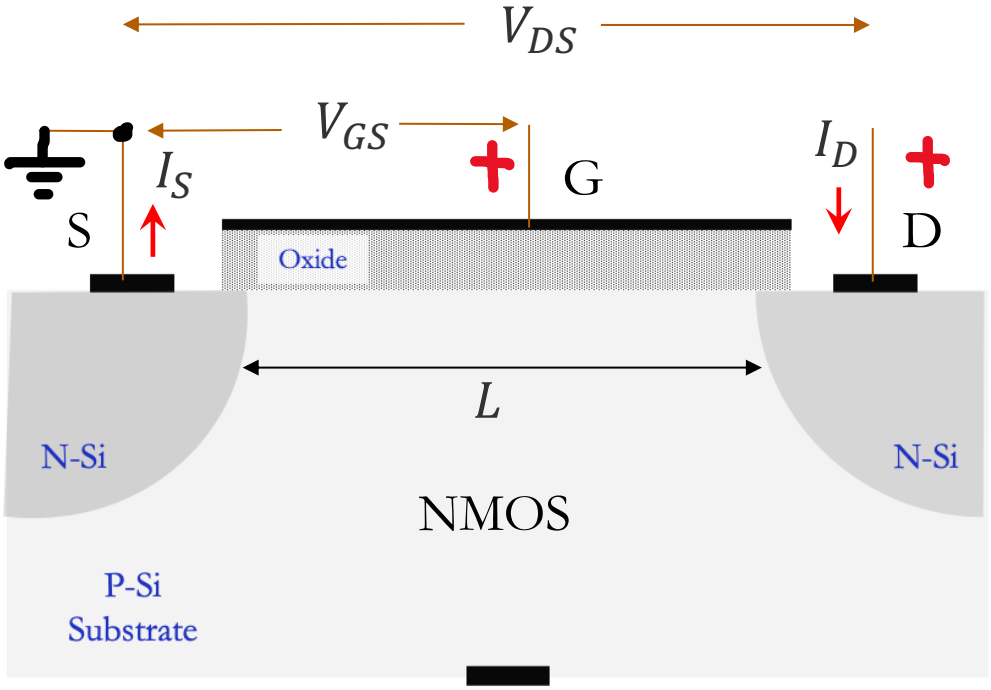
The MOSFET Structure



$$C_{ox} = \frac{\varepsilon WL}{t_{ox}}$$

NMOS Enhancement Type

Current in a MOSFET



To have any current, a channel need to be created.

$$V_{GS} \geq V_t$$

V_t is the threshold voltage. The voltage needed to start the channel. $V_{GS} < V_t$, nothing will happen $I_D = I_S = 0$. This is the Cut-off Mode.

The value of V_t (1 ~ 2 V) depends on:

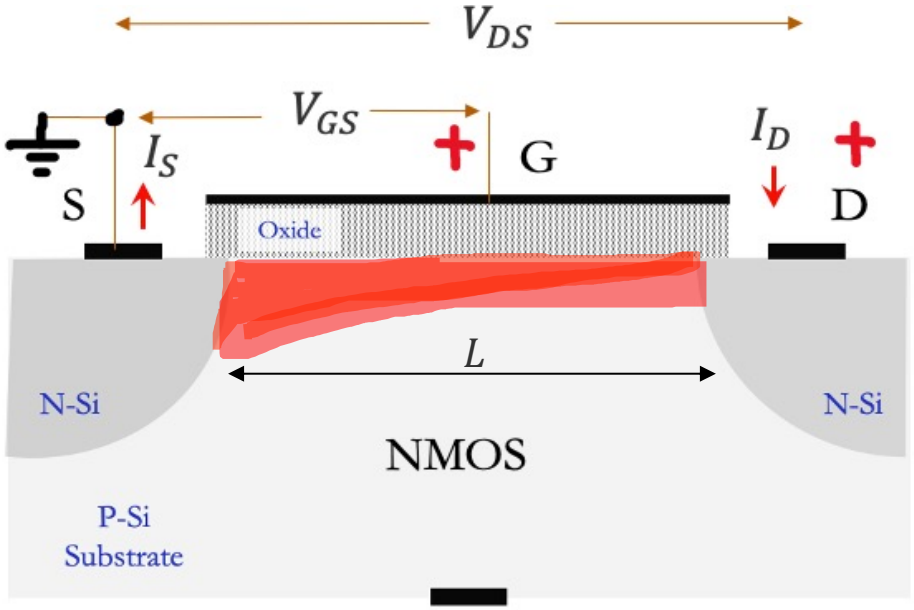
- Substrate Doping.
- Gate Oxide.
- Material.
- Body voltage.

With $V_{GS} \geq V_t$, the current depends on the value of V_{DS}.

Two regions can be identified:

1) $V_{DS} \leq V_{GS} - V_t$ (Triode)
2) $V_{DS} \geq V_{GS} - V_t$ (Saturation)

Current in a MOSFET



Electron mobility — Oxide capacitance

NMOS $$k_n = \frac{\mu_n C_{ox}}{2}\frac{W}{L}$$ → Aspect ratio (size)

PMOS $$k_p = \frac{\mu_p C_{ox}}{2}\frac{W}{L}$$

1) Cutoff Mode

$$V_{GS} < V_t$$

$$I_D = 0$$

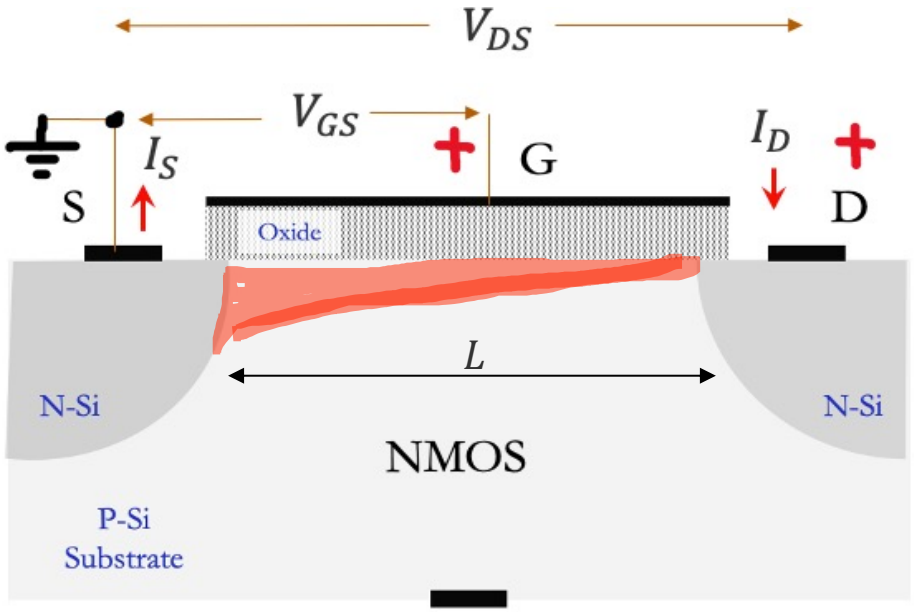
2) Triode Mode

$$V_{GS} \geq V_t \text{ and } V_{DS} \leq V_{GS} - V_t$$

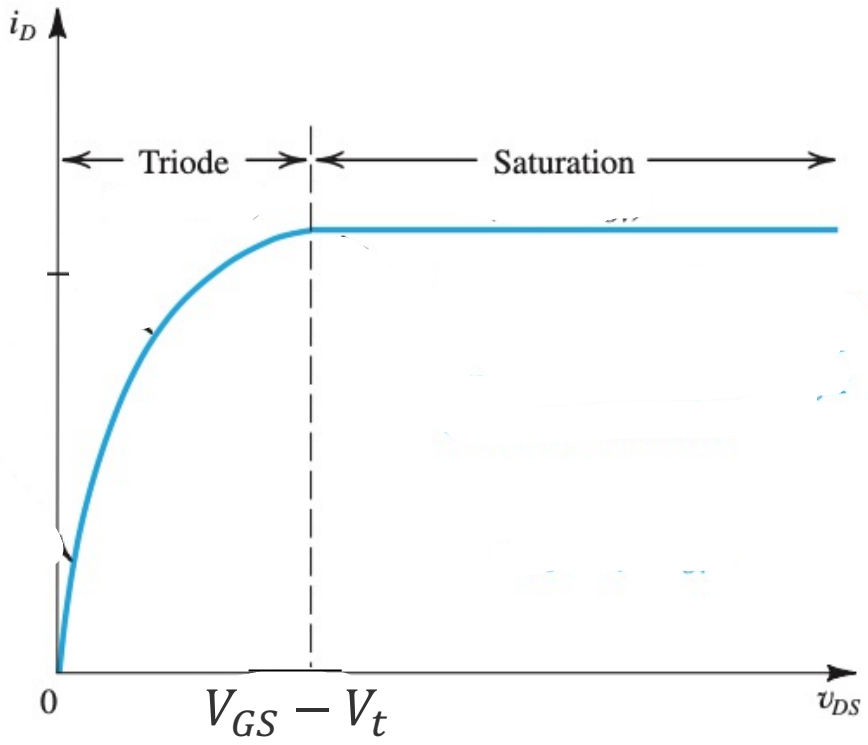
$$I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$$

$$k = \frac{\mu\, C_{ox}}{2}\frac{W}{L}$$

Current in a MOSFET



MOSFET IV Curve



Pinch-off

3) Saturation Mode

$$V_{GS} \geq V_t \text{ and } V_{DS} \geq V_{GS} - V_t$$

$$I_D = k(V_{GS} - V_t)^2$$

Summary

Mode	NMOS (V_t +ve)	PMOS (V_t -ve)
Cutoff	$V_{GS} < V_t \quad I_D = 0$	$V_{GS} > V_t \quad I_D = 0$
Triode	$V_{GS} \geq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$	$V_{GS} \leq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k\left[2(V_{GS} - V_t)V_{DS} - {V_{DS}}^2\right]$
Saturation	$V_{GS} \geq V_t$ and $V_{DS} \geq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2$	$V_{GS} \leq V_t$ and $V_{DS} \leq V_{GS} - V_t$ $I_D = k(V_{GS} - V_t)^2$

Q & A

Send me an email if
you have any
questions.